

CLASS 9 MATHEMATICS • SINDH BOARD

Factorization

Every type explained, step by step – with solved examples

$$a^2 - b^2$$

$$(a+b)^2$$

$$a^3 + b^3$$

What We'll Cover Today

1

Common Factor

$$ab + ac = a(b + c)$$

2

Grouping

4-term expressions

3

Difference of Squares

$$a^2 - b^2$$

4

Perfect Square Trinomial

$$a^2 \pm 2ab + b^2$$

5

Sum / Diff. of Cubes

$$a^3 \pm b^3$$

6

Quadratic Trinomial

$$ax^2 + bx + c$$

1

COMMON FACTOR

Type 1

When every term shares a common factor, take it outside the bracket.

$$ab + ac = a(b + c)$$

Board Tip

Always check for a common factor FIRST, before trying any other method.

SOLVED EXAMPLE

$$15a^2b - 20ab^2$$

- 1 Common factor of $15a^2b$ and $20ab^2$ is $5ab$
- 2 Divide each term by $5ab$
- 3 $15a^2b \div 5ab = 3a$, $20ab^2 \div 5ab = 4b$

$$= 5ab(3a - 4b)$$

2

GROUPING

Type 2

When there are 4 terms with no single common factor, group them in pairs.

$$ax + ay + bx + by \\ = (x + y)(a + b)$$

Board Tip

Try 2 different groupings if the first one doesn't leave a common bracket.

SOLVED EXAMPLE

$$2x^2 + 3xy - 2xz - 3yz$$

- 1 Group: $(2x^2 + 3xy) - (2xz + 3yz)$
- 2 Factor each group: $x(2x + 3y) - z(2x + 3y)$
- 3 Common bracket $(2x + 3y)$ comes out

$$= (2x + 3y)(x - z)$$

3

DIFFERENCE OF SQUARES

Type 3

$$a^2 - b^2 = (a - b)(a + b)$$

Only works with SUBTRACTION.
 $a^2 + b^2$ cannot be factorized this way.

SOLVED EXAMPLE

$$9x^2 - 25y^2$$

- 1 Write as squares: $(3x)^2 - (5y)^2$
- 2 Match to $a^2 - b^2$, so $a = 3x$, $b = 5y$
- 3 Apply the formula $(a - b)(a + b)$

$$= (3x - 5y)(3x + 5y)$$

4

PERFECT SQUARE TRINOMIAL

Type 4

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

3-Point Checklist

Is the first term a perfect square?

Is the last term a perfect square?

Does the middle term equal $2ab$?

SOLVED EXAMPLE

$$4x^2 - 12x + 9$$

- 1 $a = 2x$ (since $(2x)^2 = 4x^2$)
- 2 $b = 3$ (since $3^2 = 9$)
- 3 Check: $2ab = 2(2x)(3) = 12x$

$$= (2x - 3)^2$$

5

SUM / DIFFERENCE OF CUBES

Type 5

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Memory Trick: "SAP"**Same sign, Always Positive middle term**

SOLVED EXAMPLE

$$27x^3 - 1$$

- 1 Write as cubes: $(3x)^3 - (1)^3$
- 2 $a = 3x, b = 1$
- 3 Apply $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$= (3x - 1)(9x^2 + 3x + 1)$$

6

QUADRATIC TRINOMIAL — PART A

Type 6

Split the middle term into two numbers whose product = axc and sum = b .

$$x^2 + bx + c$$

This is the most important type for exams — spend extra practice time here.

SOLVED EXAMPLE

$$x^2 + 7x + 12$$

- 1 Need two numbers: product = 12, sum = 7
- 2 Those numbers are 3 and 4
- 3 $x^2 + 3x + 4x + 12 = x(x+3) + 4(x+3)$

$$= (x + 3)(x + 4)$$

6

QUADRATIC TRINOMIAL — PART B

When $a \neq 1$

Same idea, but now multiply $a \times c$ first — the split numbers must multiply to that.

$$ax^2 + bx + c$$

This is the board-repeated type — practice 3-4 more of these on your own.

SOLVED EXAMPLE

$$2x^2 + 5x + 3$$

- 1 $a \times c = 2 \times 3 = 6$; need product 6, sum 5 $\rightarrow$ 2 and 3
- 2 $2x^2 + 2x + 3x + 3$
- 3 $2x(x+1) + 3(x+1)$

$$= (x + 1)(2x + 3)$$

SCREENSHOT THIS

Quick Recap – All Formulas

1

Common Factor

$$ab + ac = a(b+c)$$

2

Grouping

$$ax+ay+bx+by = (x+y)(a+b)$$

3

Diff. of Squares

$$a^2-b^2 = (a-b)(a+b)$$

4

Perfect Square

$$a^2\pm 2ab+b^2 = (a\pm b)^2$$

5

Sum/Diff of Cubes

$$a^3\pm b^3 = (a\pm b)(a^2\mp ab+b^2)$$

6

Quadratic Trinomial

$$ax^2+bx+c \rightarrow \text{split middle term}$$

YOUR TURN

Practice Before You Continue

1. $8x^2 + 12x$

2. $x^2 - 49$

3. $9x^2 + 30x + 25$

4. $x^2 - 5x - 24$

5. $3x^2 + 11x + 6$

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